

LAPORAN PRAKTIKUM
PRAKTIKUM PEMROGRAMAN WEB 1
PERTEMUAN 12
DATABASE PADA PHP



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PERTEMUAN 12

DATABASE PADA PHP

A. Tujuan Percobaan

1. Membuat koneksi PHP ke database MySQL menggunakan MySQLi
2. Membangun aplikasi CRUD (Create, Read, Update, Delete) yang lengkap
3. Mengimplementasikan sistem login dan register dengan session PHP
4. Menambahkan fitur upload foto pada form data mahasiswa

B. Dasar Teori

1. Pengertian CRUD

CRUD merupakan singkatan dari Create, Read, Update, dan Delete yang merupakan operasi dasar dalam pengelolaan data pada sebuah database. Operasi Create digunakan untuk menambahkan data baru, Read digunakan untuk menampilkan data, Update digunakan untuk mengubah data yang sudah ada, dan Delete digunakan untuk menghapus data dari database. Konsep CRUD banyak digunakan dalam pengembangan aplikasi berbasis web maupun desktop karena mempermudah proses manipulasi data secara sistematis.

2. PHP (Hypertext Preprocessor)

PHP adalah bahasa pemrograman server-side yang digunakan untuk membangun aplikasi web dinamis. PHP dapat disisipkan ke dalam kode HTML dan dijalankan pada server untuk menghasilkan halaman web yang interaktif. PHP banyak digunakan karena mudah dipelajari, mendukung koneksi database, dan kompatibel dengan berbagai sistem operasi maupun web server.

Dalam praktikum ini, PHP digunakan untuk:

- Menghubungkan aplikasi dengan database MySQL
- Memproses data formulir
- Menampilkan data mahasiswa
- Melakukan proses tambah, ubah, dan hapus data
- Mengelola session login pengguna

3. MySQL

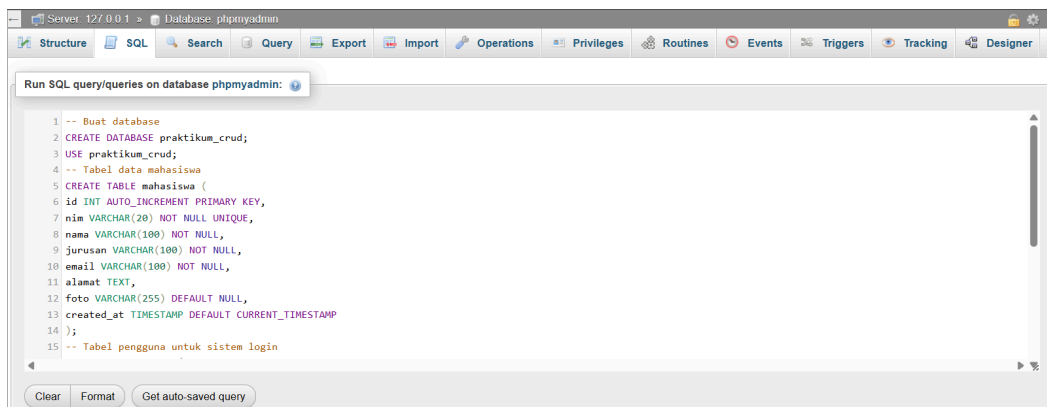
MySQL merupakan sistem manajemen basis data relasional (Relational Database Management System / RDBMS) yang digunakan untuk menyimpan dan mengelola data secara terstruktur. MySQL menggunakan bahasa SQL (Structured Query Language) untuk menjalankan perintah pengolahan data seperti SELECT, INSERT, UPDATE, dan DELETE.

4. XAMPP

XAMPP adalah paket perangkat lunak yang berisi Apache, MySQL, PHP, dan phpMyAdmin yang digunakan untuk membangun server lokal (localhost). XAMPP memudahkan proses pengembangan aplikasi web karena seluruh komponen server dapat dijalankan dalam satu aplikasi.

C. Hasil dan Pembahasan

1. Membuat Database MySQL



Gambar 1. Tampilan membuat database pada Xampp

2. config.php

```
<?php
session_start();
// Konfigurasi koneksi database
$host      = "localhost";
$username  = "root";
$password  = "";
$database  = "praktikum_crud";
// Buat koneksi
$conn = mysqli_connect($host, $username, $password, $database);
// Cek koneksi
if (!$conn) {
    die("Koneksi gagal: " . mysqli_connect_error());
}
```

```
// Cek apakah user sudah login
function isLoggedIn() {
    return isset($_SESSION['user_id']);
}
// Redirect ke login jika belum login
function requireLogin() {
    if (!isLoggedIn()) {
        header("Location: login.php");
        exit();
    }
}
// Fungsi upload foto
function uploadFile($file, $target_dir = "uploads/mahasiswa/") {
    // Buat folder jika belum ada
    if (!file_exists($target_dir)) {
        mkdir($target_dir, 0777, true);
    }
    $imageFileType = strtolower(pathinfo($file["name"], PATHINFO_EXTENSION));
    // Validasi ukuran (max 5MB)
    if ($file["size"] > 5000000) {
        return ['success' => false, 'message' => 'File terlalu besar. Maks 5MB.'];
    }
    // Validasi format
    $allowed = ["jpg", "jpeg", "png", "gif"];
    if (!in_array($imageFileType, $allowed)) {
        return ['success' => false, 'message' => 'Format tidak didukung.'];
    }
    // Generate nama file unik agar tidak tabrakan
    $new_filename = uniqid() . "." . $imageFileType;
    $target_file = $target_dir . $new_filename;
    if (move_uploaded_file($file["tmp_name"], $target_file)) {
        return ['success' => true, 'filename' => $new_filename];
    }
    return ['success' => false, 'message' => 'Gagal upload file.'];
}
// Fungsi hapus file foto
function deleteFile($filename, $dir = "uploads/mahasiswa/") {
    if ($filename && file_exists($dir . $filename)) {
        unlink($dir . $filename);
    }
}
?>
```

3. login.php

```
<?php
include_once("config.php");
if (isLoggedIn()) { header('Location: index.php'); exit(); }
$error = ''; $success = isset($_GET['registered']) ? 'Registrasi berhasil. Silakan login.' : '';

if (isset($_POST['login'])) {
    $username = mysqli_real_escape_string($conn, $_POST['username'] ?? '');
    $res = mysqli_query($conn, "SELECT * FROM users WHERE username = '$username' OR email = '$username'");
    if ($res && mysqli_num_rows($res) == 1 && password_verify($_POST['password'] ?? '', ($user = mysqli_fetch_assoc($res))['password'])) {
        $_SESSION['user_id'] = $user['id']; $_SESSION['username'] = $user['username'];
        $_SESSION['full_name'] = $user['full_name'];
        header('Location: index.php'); exit();
    }
    $error = 'Username atau password salah!';
}
?>
```

```

<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8"><title>Login</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light vh-100 d-flex align-items-center justify-content-center">
<div class="card p-4 shadow-sm border-0 w-100" style="max-width: 380px;">
    <h5 class="text-center mb-3 fw-bold">Masuk Sistem</h5>
    <?php if ($error) echo '<div class="alert alert-danger p-2 small
text-center">'. $error. '</div>'; ?>
    <?php if ($success) echo '<div class="alert alert-success p-2 small
text-center">'. $success. '</div>'; ?>

    <form method="POST">
        <div class="mb-2"><label class="form-label small mb-1">Username / Email</label><input
type="text" name="username" class="form-control form-control-sm" required
autocomplete="off"></div>
        <div class="mb-3"><label class="form-label small mb-1">Password</label><input
type="password" name="password" class="form-control form-control-sm" required></div>
        <button type="submit" name="login" class="btn btn-primary btn-sm w-100
fw-medium">Login</button>
    </form>
    <div class="text-center small text-muted mt-3">Belum punya akun? <a href="register.php"
class="text-decoration-none">Daftar Akun</a></div>
</div>
</body>
</html>

```

4. register.php

```

<?php
include_once("config.php");
if (isLoggedIn()) { header('Location: index.php'); exit(); }
$errors = [];

if (isset($_POST['register'])) {
    foreach (['full_name', 'username', 'email'] as $f) $$f = mysqli_real_escape_string($conn,
trim($_POST[$f] ?? ''));
    $password = $_POST['password'] ?? ''; $confirm = $_POST['confirm_password'] ?? '';

    if (!$full_name || !$username || !$email || !$password) $errors[] = 'Semua kolom wajib
diisi!';
    if ($password !== $confirm) $errors[] = 'Konfirmasi password tidak cocok!';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = 'Email tidak valid!';
    if (empty($errors) && mysqli_num_rows(mysqli_query($conn, "SELECT id FROM users WHERE
username='$username' OR email='$email'")) > 0) $errors[] = 'Username atau email sudah
terdaftar!';
}

```

```

        if (empty($errors) && mysqli_query($conn, "INSERT INTO users (full_name, username, email,
password) VALUES ('$full_name', '$username', '$email', '" . password_hash($password,
PASSWORD_DEFAULT) . "')")) {
            header('Location: login.php?registered=1'); exit();
        }
    }
}
?>
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8"><title>Daftar Akun</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light vh-100 d-flex align-items-center justify-content-center">
<div class="card p-4 shadow-sm border-0 w-100" style="max-width: 380px;">
    <h3 class="text-center mb-3 fw-bold">Daftar Akun</h3>
    <?php if ($errors) echo '<div class="alert alert-danger p-2 small">'.implode('<br>',
$errors).'</div>'; ?>

    <form method="POST">
        <?php foreach(['full_name'=>'Nama Lengkap', 'username'=>'Username', 'email'=>'Email']
as $k => $v): ?>
            <div class="mb-2"><label class="form-label small mb-1"><?= $v ?></label><input
type="<?= $k=='email'? 'email': 'text' ?>" name="<?= $k ?>" class="form-control form-control-sm"
value="<?= htmlspecialchars($_POST[$k] ?? '') ?>" required></div>
            <?php endforeach; ?>
            <div class="mb-2"><label class="form-label small mb-1">Password</label><input
type="password" name="password" class="form-control form-control-sm" required></div>
            <div class="mb-3"><label class="form-label small mb-1">Konfirmasi
Password</label><input type="password" name="confirm_password" class="form-control
form-control-sm" required></div>
            <button type="submit" name="register" class="btn btn-primary btn-sm w-100
fw-medium">Register</button>
        </form>
        <div class="text-center small text-muted mt-3">Sudah punya akun? <a href="login.php"
class="text-decoration-none">Login di sini</a></div>
    </div>
</body>
</html>

```

5. index.php

```

<?php
include_once("config.php"); requireLogin();
$result = mysqli_query($conn, "SELECT * FROM mahasiswa ORDER BY id DESC");
?>
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <title>Dashboard</title>

```



```

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="container py-4" style="max-width: 800px;">

    <div class="d-flex justify-content-between align-items-center mb-3">
        <h6 class="m-0">Halo, <?= htmlspecialchars($_SESSION['full_name']) ?? '' ?></h6>
        <a href="logout.php" class="btn btn-outline-danger btn-sm" onclick="return
confirm('Keluar?')">Logout</a>
    </div>

    <a href="tambah.php" class="btn btn-primary btn-sm mb-3">+ Tambah Mahasiswa</a>

    <?php if (isset($_SESSION['message'])): ?>
        <div class="alert alert-success p-2 small"><?= $_SESSION['message'];
unset($_SESSION['message']); ?></div>
    <?php endif; ?>

    <table class="table table-bordered table-hover table-sm small align-middle">
        <thead class="table-light">
            <tr><th>Foto</th><th>NIM</th><th>Nama</th><th>Jurusan</th><th>Aksi</th></tr>
        </thead>
        <tbody>
            <?php while ($row = mysqli_fetch_assoc($result)): ?>
                <tr>
                    <td>
                        <?php if($row['foto']): ?>
                        <?php else: ?> N/A <?php endif; ?>
                    </td>
                    <td class="fw-bold"><?= htmlspecialchars($row['nim']) ?></td>
                    <td><?= htmlspecialchars($row['nama']) ?></td>
                    <td><?= htmlspecialchars($row['jurusan']) ?></td>
                    <td>
                        <a href="edit.php?id=<?= $row['id'] ?>" class="btn btn-warning btn-sm py-0
px-2 text-white">Edit</a>
                        <a href="hapus.php?id=<?= $row['id'] ?>" class="btn btn-danger btn-sm py-0
px-2" onclick="return confirm('Hapus?')">Hapus</a>
                    </td>
                </tr>
            <?php endwhile; ?>
        </tbody>
    </table>

</body>
</html>

```

6. tambah.php

```

<?php
include_once("config.php"); requireLogin();
$errors = []; $success = '';

```

```

if (isset($_POST['submit'])) {
    foreach (['nim', 'nama', 'jurusan', 'email', 'alamat'] as $f) $$f =
mysqli_real_escape_string($conn, $_POST[$f] ?? '');

    if (!$nim || !$nama || !$jurusan || !$email) $errors[] = 'Semua kolom * wajib diisi!';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = 'Format email tidak valid!';
    if (mysqli_num_rows(mysqli_query($conn, "SELECT nim FROM mahasiswa WHERE nim='$nim'")))
$errors[] = 'NIM sudah terdaftar!';

    $foto = null;
    if (!empty($_FILES['foto']['name'])) {
        $upload = uploadFile($_FILES['foto']);
        $upload['success'] ? $foto = $upload['filename'] : $errors[] = $upload['message'];
    }

    if (empty($errors)) {
        $foto_sql = $foto ? "'$foto'" : 'NULL';
        if (mysqli_query($conn, "INSERT INTO mahasiswa (nim, nama, jurusan, email, alamat,
foto) VALUES ('$nim','$nama','$jurusan','$email','$alamat',$foto_sql)")) { $success = 'Data
berhasil ditambahkan!'; $_POST = []; }
        else { $errors[] = 'Gagal menyimpan data.'; if ($foto) deleteFile($foto); }
    }
}
?>
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8"><title>Tambah Mahasiswa</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="bg-light py-5">
<div class="container" style="max-width: 450px;">
    <a href="index.php" class="text-decoration-none small text-secondary d-inline-block mb-2
fw-medium">← Kembali ke Dashboard</a>
    <div class="card p-3 shadow-sm border-0">
        <h5 class="text-center mb-3 fw-bold">Tambah Mahasiswa</h5>
        <?php if ($errors) echo '<div class="alert alert-danger p-2 small">'.implode('<br>',
$errors).'

```

```

        <div class="mb-3"><label class="form-label small mb-1">Alamat</label><textarea
name="alamat" class="form-control form-control-sm" rows="2"><?=
htmlspecialchars($_POST['alamat'] ?? '') ?></textarea></div>

        <div class="d-flex gap-2"><button type="submit" name="submit" class="btn
btn-primary btn-sm flex-grow-1">Simpan</button><a href="index.php" class="btn btn-secondary
btn-sm px-3">Batal</a></div>

    </form>
</div>
</div>
</body>
</html>

```

7. hapus.php

```

<?php
include_once("config.php");
requireLogin();

if (isset($_GET['id'])) {
    $id = (int)$_GET['id'];

    // Ambil nama foto SEBELUM data dihapus
    $result = mysqli_query($conn, "SELECT foto FROM mahasiswa WHERE id=$id");
    if (mysqli_num_rows($result) > 0) {
        $row = mysqli_fetch_assoc($result);
        $foto = $row["foto"];

        // Hapus data dari database
        if (mysqli_query($conn, "DELETE FROM mahasiswa WHERE id=$id")) {
            // Hapus file foto jika ada
            if ($foto) deleteFile($foto);
            $_SESSION['message'] = 'Data berhasil dihapus!';
        } else {
            $_SESSION['error'] = 'Error: ' . mysqli_error($conn);
        }
    }
}

// Kembali ke halaman utama
header('Location: index.php');
exit();
?>

```

8. edit.php

```

<?php
include_once("config.php"); requireLogin();
$id = (int)($_GET["id"] ?? 0);
$row = mysqli_fetch_assoc(mysqli_query($conn, "SELECT * FROM mahasiswa WHERE id=$id"));
if (!$row) { header('Location: index.php'); exit(); }
$errors = [];

if (isset($_POST['update'])) {

```

```

        foreach (['nim', 'nama', 'jurusan', 'email', 'alamat'] as $f) $$f =
mysqli_real_escape_string($conn, $_POST[$f] ?? '');
        if (!$nim || !$nama || !$jurusan || !$email) $errors[] = 'Semua kolom * wajib diisi!';
        if (mysqli_num_rows(mysqli_query($conn, "SELECT id FROM mahasiswa WHERE nim='$nim' AND id
!= $id"))) $errors[] = 'NIM sudah terdaftar!';

        $foto = $row['foto'];
        if (!empty($_FILES['foto']['name'])) {
            $up = uploadFile($_FILES['foto']);
            if ($up['success']) { if($foto) deleteFile($foto); $foto = $up['filename']; } else
$errors[] = $up['message'];
        }
        if ($_POST['hapus_foto'] ?? '') == '1' { if($foto) deleteFile($foto); $foto = null; }

        if (empty($errors) && mysqli_query($conn, "UPDATE mahasiswa SET nim='$nim', nama='$nama',
jurusan='$jurusan', email='$email', alamat='$alamat', foto=" . ($foto ? "'$foto'" : "NULL") .
" WHERE id=$id")) {
            $_SESSION['message'] = 'Data diperbarui!'; header('Location: index.php'); exit();
        }
    }
}
?>
<!DOCTYPE html>
<html lang="id">
<head><meta charset="UTF-8"><title>Edit</title><link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet"></head>
<body class="bg-light py-4">
<div class="container" style="max-width: 400px;"><div class="card p-3 shadow-sm border-0">
    <h5 class="text-center mb-3 fw-bold">Edit Mahasiswa</h5>
    <?php if ($errors) echo '<div class="alert alert-danger p-2 small">'.implode('<br>',
$errors).'

```

```

    </form>
</div></div>
</body>
</html>

```

9. detail.php

```

<?php
include_once("config.php"); requireLogin();
$id = (int)($_GET['id'] ?? 0);
$result = mysqli_query($conn, "SELECT * FROM mahasiswa WHERE id = $id");
$row = mysqli_fetch_assoc($result);
if (!$row) { header("Location: index.php"); exit(); }
?>

<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8"><title>Detail</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body class="container py-5" style="max-width: 400px;">
    <div class="card p-3 border-0 shadow-sm">
        <h5 class="fw-bold text-center mb-3">Detail Mahasiswa</h5>
        <div class="text-center mb-3"><?= $row['foto'] ? '' : 'N/A' ?></div>
        <div class="small mb-3">
            <p class="mb-1"><b>NIM:</b> <?= htmlspecialchars($row['nim']) ?></p>
            <p class="mb-1"><b>Nama:</b> <?= htmlspecialchars($row['nama']) ?></p>
            <p class="mb-1"><b>Jurusan:</b> <?= htmlspecialchars($row['jurusan']) ?></p>
            <p class="mb-1"><b>Email:</b> <?= htmlspecialchars($row['email']) ?></p>
            <p class="mb-1"><b>Alamat:</b> <?= htmlspecialchars($row['alamat']) ?></p>
        </div>
        <div class="d-flex gap-2">
            <a href="index.php" class="btn btn-secondary btn-sm w-100">Kembali</a>
            <a href="edit.php?id=<?= $row['id'] ?>" class="btn btn-warning btn-sm text-white
w-100">Edit</a>
        </div>
    </div>
</body>
</html>

```

10. logout.php

```

<?php
session_start();

// Hapus semua data sesi
session_unset();
session_destroy();

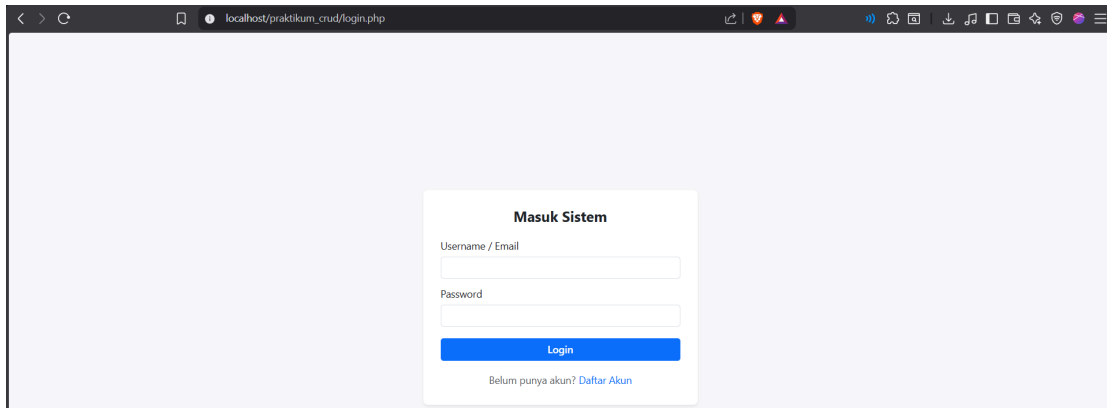
// Redirect ke halaman login
header('Location: login.php');

```

```
exit();  
?>
```

11. Menguji website yang telah dibuat

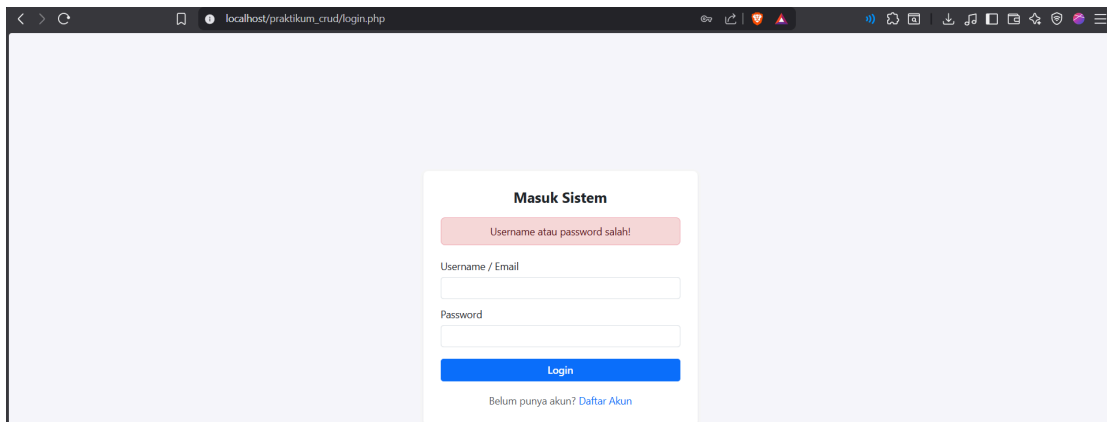
- a. Untuk masuk ke sistem kita perlu melakukan proses login



The screenshot shows a web browser window with the address bar displaying 'localhost/praktikum_crud/login.php'. The main content area features a light purple background. In the center, there is a white rectangular form titled 'Masuk Sistem'. Inside the form, there are two input fields: 'Username / Email' and 'Password'. Below these fields is a blue button labeled 'Login'. At the bottom of the form, there is a link that says 'Belum punya akun? [Daftar Akun](#)'.

Gambar 2. Tampilan login.php

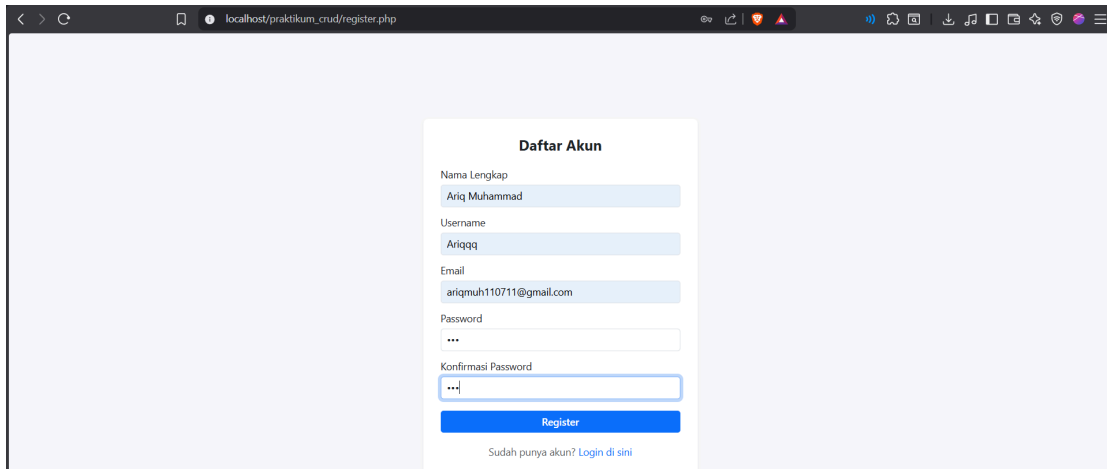
- b. Tampilan jika password/username salah saat melakukan login



The screenshot shows the same web browser window as before, but now the login form is displaying an error message. A red rectangular box with the text 'Username atau password salah!' is positioned above the 'Username / Email' input field. The rest of the form, including the 'Password' field, the 'Login' button, and the 'Daftar Akun' link, remains visible below the error message.

Gambar 3. Tampilan error pada login.php

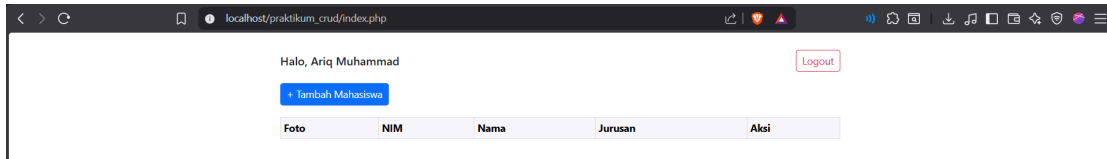
- c. Jika tidak ada akun sebelumnya, kita melakukan registrasi akun baru



The screenshot shows a web browser window with the address bar displaying 'localhost/praktikum_crud/register.php'. The main content area features a light purple background. In the center, there is a white rectangular form titled 'Daftar Akun'. Inside the form, there are five input fields: 'Nama Lengkap' (containing 'Ariq Muhammad'), 'Username' (containing 'Ariqqq'), 'Email' (containing 'ariqmuh110711@gmail.com'), 'Password' (containing '***'), and 'Konfirmasi Password' (containing '***'). Below these fields is a blue button labeled 'Register'. At the bottom of the form, there is a link that says 'Sudah punya akun? [Login di sini](#)'.

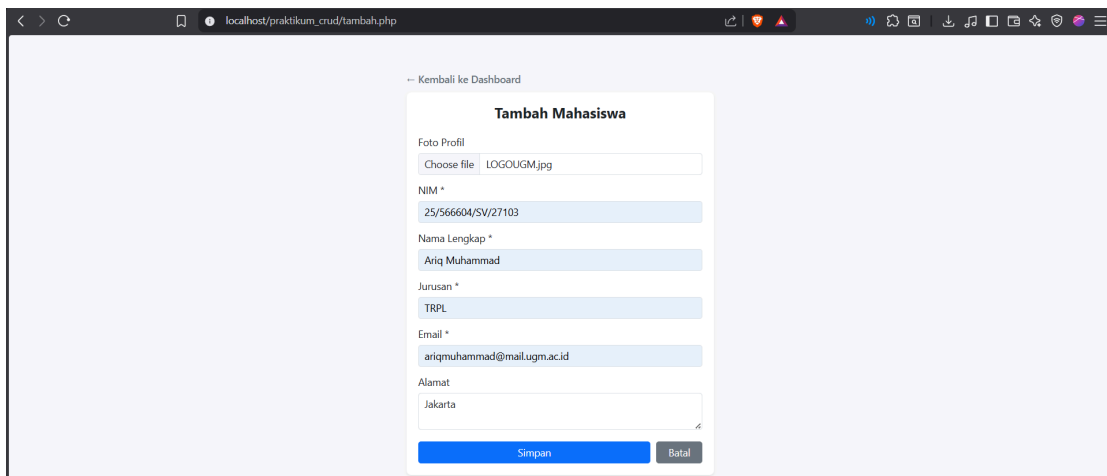
Gambar 4. Tampilan register.php

- d. Jika sudah melakukan register/login akan masuk ke index.php dan tampilannya seperti ini:



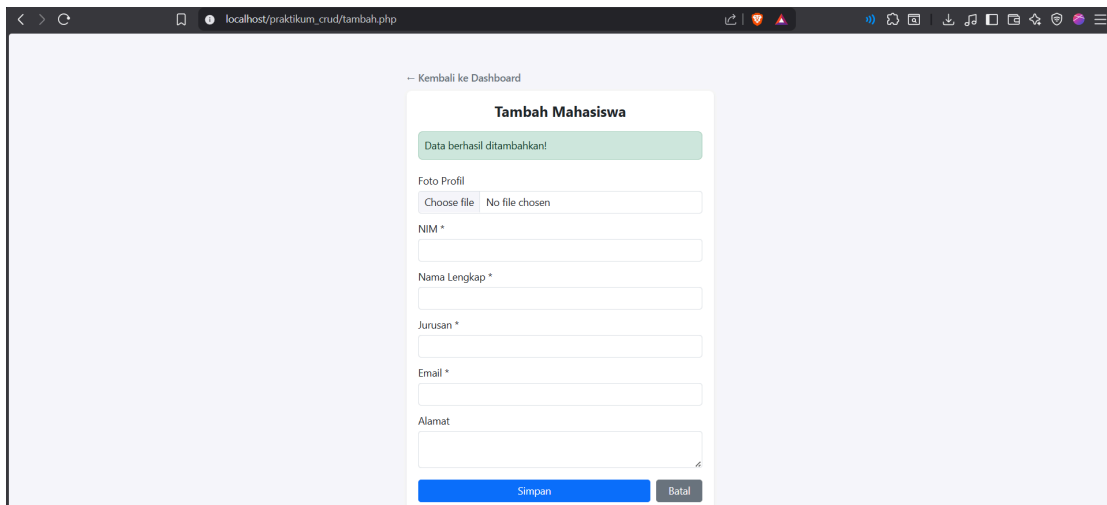
Gambar 5. Tampilan index.php

- e. Setelah menekan tombol tambah mahasiswa akan ke halaman tambah.php dimana kita mengisi biodata mahasiswa yang akan masuk ke database.



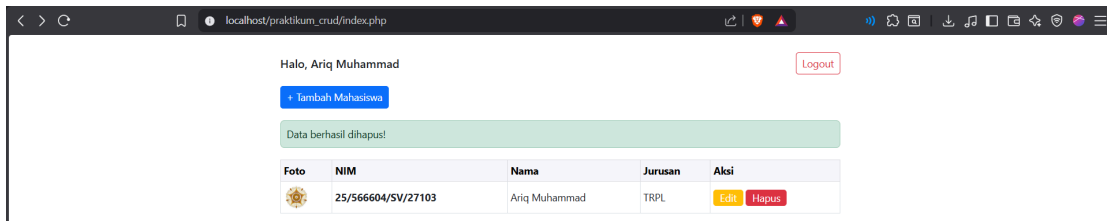
Gambar 6. Tampilan tambah.php

- f. Tampilan setelah data berhasil disimpan



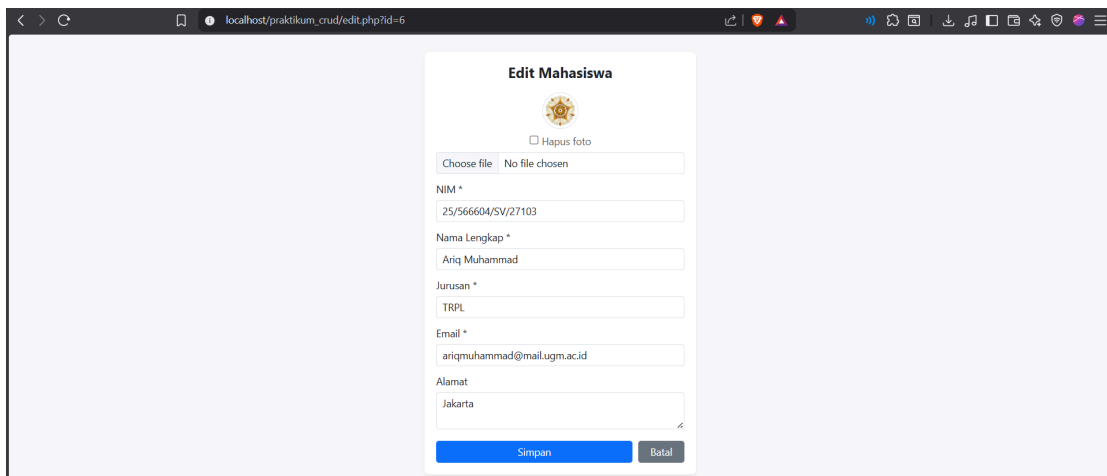
Gambar 7. Tampilan berhasil menambah data pada tambah.php

g. Data yang sudah disimpan akan muncul pada index.php



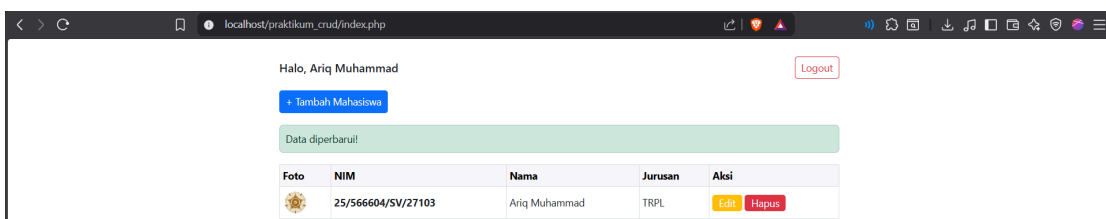
Gambar 8. Tampilan index.php saat berhasil

h. Jika ada data yang salah, dapat kita ubah dengan menekan tombol edit yang akan menuju ke halaman edit.php



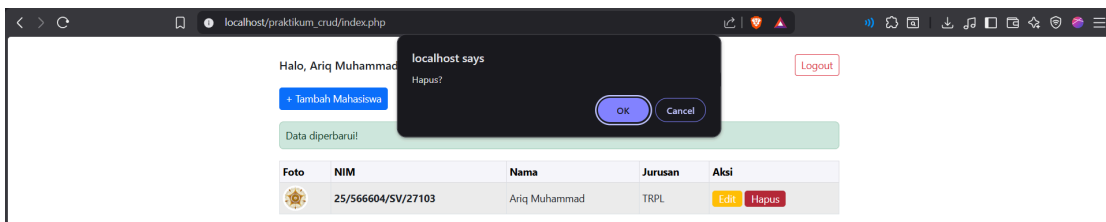
Gambar 9. Tampilan edit.php

i. Tampilan halaman index.php setelah data berhasil di edit/update



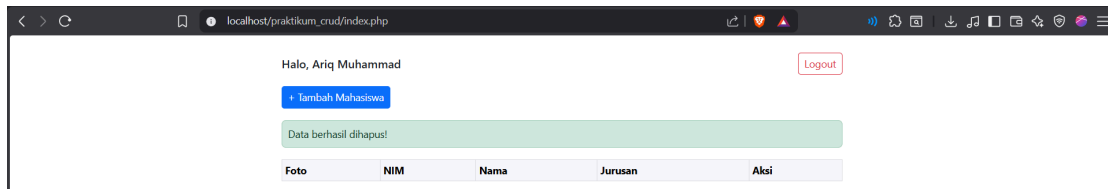
Gambar 10. Tampilan berhasil mengedit data

j. Jika ingin menghapus data kita tekan tombol hapus yang akan menampilkan



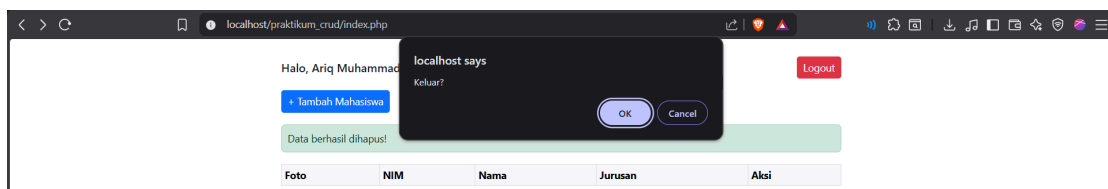
Gambar 11. Tampilan jika menghapus data

k. Data berhasil dihapus



Gambar 11. Tampilan data berhasil dihapus

l. Tampilan saat ingin logout



Gambar 12. Tampilan saat logout

D. Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa aplikasi CRUD (Create, Read, Update, Delete) berbasis PHP dan MySQL berhasil dibuat dan dijalankan dengan baik pada server lokal menggunakan XAMPP. Sistem mampu melakukan pengelolaan data mahasiswa seperti menambahkan, menampilkan, mengubah, dan menghapus data secara dinamis melalui database MySQL.

Selain itu, fitur tambahan berupa halaman detail mahasiswa berhasil diimplementasikan untuk menampilkan informasi lengkap mahasiswa beserta foto dengan ukuran yang lebih besar. Validasi input NIM juga berhasil diterapkan pada halaman tambah dan edit data sehingga NIM hanya dapat diisi menggunakan angka dengan panjang 8 hingga 12 digit. Validasi tersebut membantu menjaga konsistensi dan keakuratan data yang disimpan di dalam database.

Dari praktikum ini dapat dipahami bagaimana cara menghubungkan PHP dengan database MySQL, mengelola data menggunakan query SQL, menerapkan validasi input, serta membangun antarmuka aplikasi web sederhana yang interaktif dan terstruktur.

E. Daftar Pustaka

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